

■ 1.2.4 Manipulating Elements of Lists

When you are doing calculations, you most often want to treat a whole list as a single object. Sometimes, however, you need to pick individual elements of the list.

The elements in *Mathematica* lists are numbered in order. The first element has “index” 1. In this respect, lists in *Mathematica* work very much like “arrays” in other computer languages.

$\{a, b, c\}$	a list
<code>list[[i]]</code>	the i^{th} element of <i>list</i> (the first element is <code>list[[1]]</code>)
<code>list[{i, j, ...}]</code>	a list of the $i^{\text{th}}, j^{\text{th}}, \dots$ elements of <i>list</i>
<code>list[[i]] = value</code>	reset the i^{th} element of <i>list</i>
<code>list[[i, j, ...]]</code>	an element of a nested list

Operations on list elements.

Here is a list of four elements.	<code>In[1]:= {5, 1, 6, 2}</code> <code>Out[1]= {5, 1, 6, 2}</code>
This extracts the third element of the list.	<code>In[2]:= %[[3]]</code> <code>Out[2]= 6</code>
Here is a list of numbers, assigned the name <code>w</code> .	<code>In[3]:= w = Range[0, 2, 0.4]</code> <code>Out[3]= {0, 0.4, 0.8, 1.2, 1.6, 2.}</code>
This picks out the second element of <code>w</code> .	<code>In[4]:= w[[2]]</code> <code>Out[4]= 0.4</code>
This gives elements 2, 3 and 4.	<code>In[5]:= w[[{2,3,4}]]</code> <code>Out[5]= {0.4, 0.8, 1.2}</code>
You can use <code>Range</code> to specify the indices for <i>ranges</i> of elements.	<code>In[6]:= w[[Range[2,5]]]</code> <code>Out[6]= {0.4, 0.8, 1.2, 1.6}</code>
This resets the second element of <code>w</code> .	<code>In[7]:= w[[2]] = 10</code> <code>Out[7]= 10</code>

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If you ask for `w`, you will see that it has now been modified.

```
In[8]:= w  
Out[8]= {0, 10, 0.8, 1.2, 1.6, 2.}
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